HOMEWORK 8.1 · UNIT 8 · NC.8.G.3 · NC.8.G.2

Transformations (Intro)

Apr 6 / Apr 7 · Targets: G.3-2, G.3-3, G.2-2, G.2-3, G.3-6, G.3-7, G.2-7, G.2-8

Name _____ Date _____ Block _____

Remember: Translation (slide): $(x + a, y + b)$. **Reflection** over the x-axis: $(x, -y)$; over the y-axis: $(-x, y)$. These moves are **rigid**: same size and shape, so the image ≅ the pre-image. Name image points with primes: A → A′.
Example: $(3, 2)$ translated 1 right and 4 down → $(3 + 1, 2 - 4) = (4, -2)$.

SCORE 3.0 Approaching · the basics

Targets: G.3-2, G.3-3, G.2-2, G.2-3

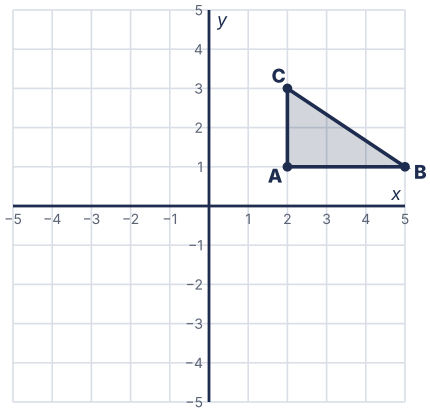
1 Translate G(−3, 7) using the rule $(x + 6, y - 5)$. Find G′. G.3-2

Ⓐ (3, 2) Ⓑ (3, 12) Ⓒ (−9, 12) Ⓓ (3, −2)

2 Reflect N(−6, 2) over the y-axis. Find N′. G.3-3

Ⓐ (−6, −2) Ⓑ (6, −2) Ⓒ (6, 2) Ⓓ (2, −6)

3 Reflect △ABC over the x-axis. A(2, 1), B(5, 1), C(2, 3). Write the coordinates of A′, B′, and C′. Draw the image. G.2-2



4 Write the translation rule for: 4 units left and 1 unit up. G.3-2

Hint: Use the form $(x + a, y + b)$.

5 Translations and reflections keep the same size and shape. G.2-3

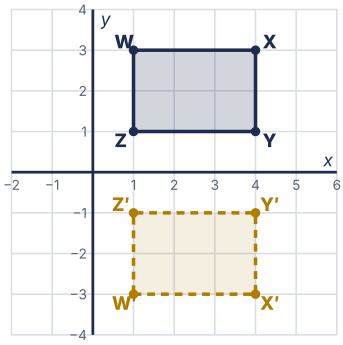
True False

SCORE 4.0 Proficient · on target

Targets: G.3-6, G.3-7

G.3-6

6 Which transformation maps rectangle WXYZ onto W′X′Y′Z′?



(A) Reflection over the *y*-axis
(B) Translation 6 down
(C) Reflection over the *x*-axis
(D) Rotation 180°

G.3-7

7 **Ava** made a mistake. Circle it. Explain it. Fix it.

AVA'S WORK

Translate Q(2, −3) by $(x - 5, y + 4)$.

Q′ = (2 + 5, −3 + 4) = (7, 1)

G.3-6

8 After the rule $(x + 5, y - 3)$, the image is Q′(4, −1). Find the original point Q.

Hint: Undo the rule: subtract 5 from x, add 3 to y.

SCORE 5.0 Mastery · go deeper

Targets: G.2-7, G.2-8

G.2-7

9 A figure is reflected over the *y*-axis. Then it is translated 2 units down. Is the final image congruent to the original? Explain.

The image _____ congruent because _____ .

G.2-8

10 Kofi says, "To move a point from (3, 4) to (3, 1), I can translate it 3 units down." Rosa says, "I can reflect it over the *x*-axis." Who is right? Explain.

_____ is right because _____ .